
School Management System

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Report

1. System Introduction

The School Management System is a RESTful API built using Spring Boot and Spring Data JPA. It is designed to manage the core entities of an educational institution: Teachers, Addresses, Courses, and Students.

2. API Endpoints Summary

The system exposes 23 REST endpoints across four controllers, running on localhost:8080.

Method	Endpoint	Description
GET	/api/v1/teacher/get	Get all teachers
POST	/api/v1/teacher/add	Add new teacher
PUT	/api/v1/teacher/update/{id}	Update teacher
DELETE	/api/v1/teacher/delete/{id}	Delete teacher
GET	/api/v1/teacher/details/{id}	Get teacher full details
GET	/api/v1/address/get/{id}	Get teacher address
POST	/api/v1/address/add/{teacherId}	Add teacher address
PUT	/api/v1/address/update/{id}	Update teacher address
DELETE	/api/v1/address/delete/{id}	Delete teacher address
GET	/api/v1/course/get	Get all courses
POST	/api/v1/course/add	Add new course
PUT	/api/v1/course/update/{id}	Update course
DELETE	/api/v1/course/delete/{id}	Delete course
PUT	/api/v1/course/assign/{courseId}/{teacherId}	Assign course to teacher
GET	/api/v1/course/teacher-name/{courseId}	Get teacher name by course
GET	/api/v1/course/students/{courseId}	Get student list in course
GET	/api/v1/student/get	Get all students
POST	/api/v1/student/add	Add new student
PUT	/api/v1/student/update/{id}	Update student
DELETE	/api/v1/student/delete/{id}	Delete student
PUT	/api/v1/student/assign/{studentId}/{courseId}	Register student in course
PUT	/api/v1/student/change-major/{studentId}/{major}	Change student major

3. Postman Test Results

All endpoints were tested using Postman. The following screenshots demonstrate the successful execution of key API operations with 200 OK responses.

3.1 Get Teacher Details

Endpoint:

GET /api/v1/teacher/details/1

Returns the full details of a teacher including their address and list of assigned courses.

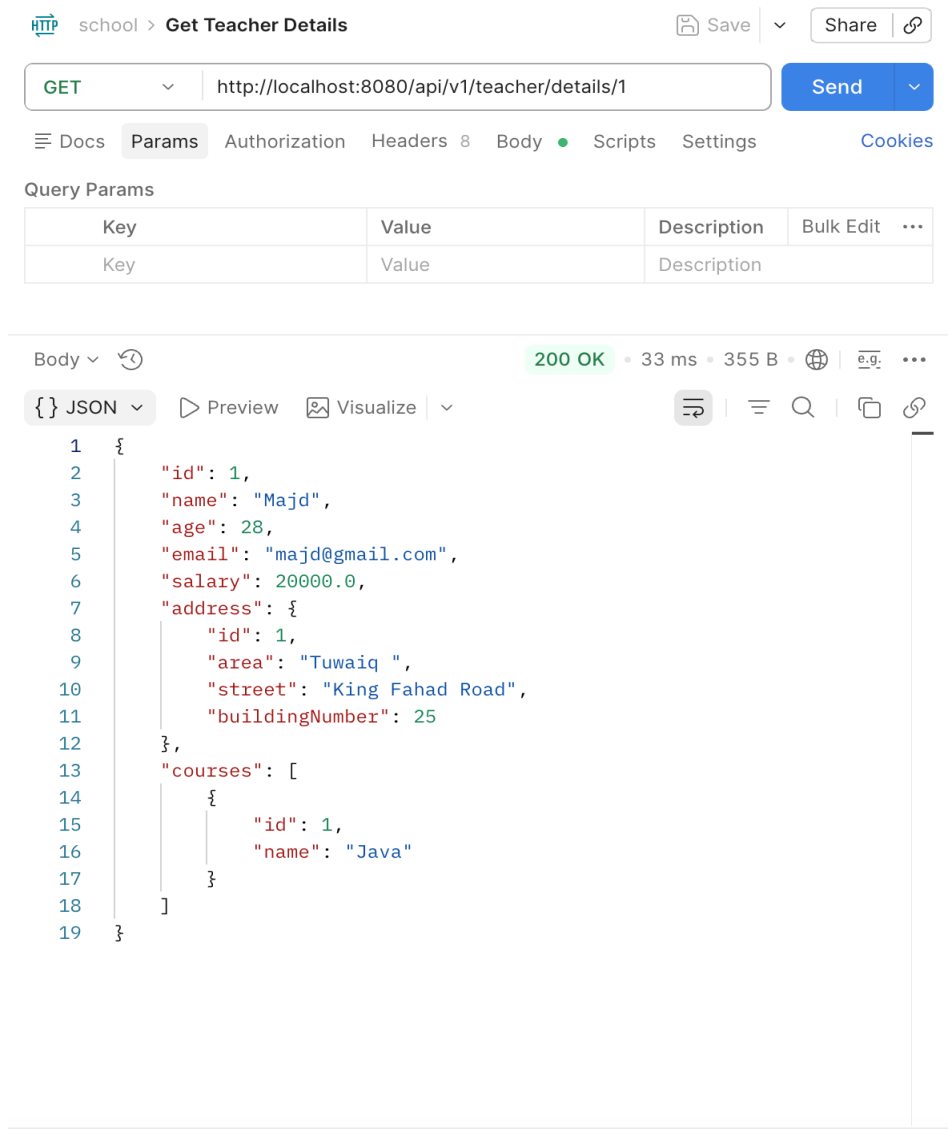


Figure 1 - Get Teacher Details: returns id, name, age, email, salary, address, and courses

3.2 Get Teacher Name by Course

Endpoint:

GET /api/v1/course/teacher-name/1

Returns the name of the teacher assigned to a specific course. Throws an exception if no teacher has been assigned.

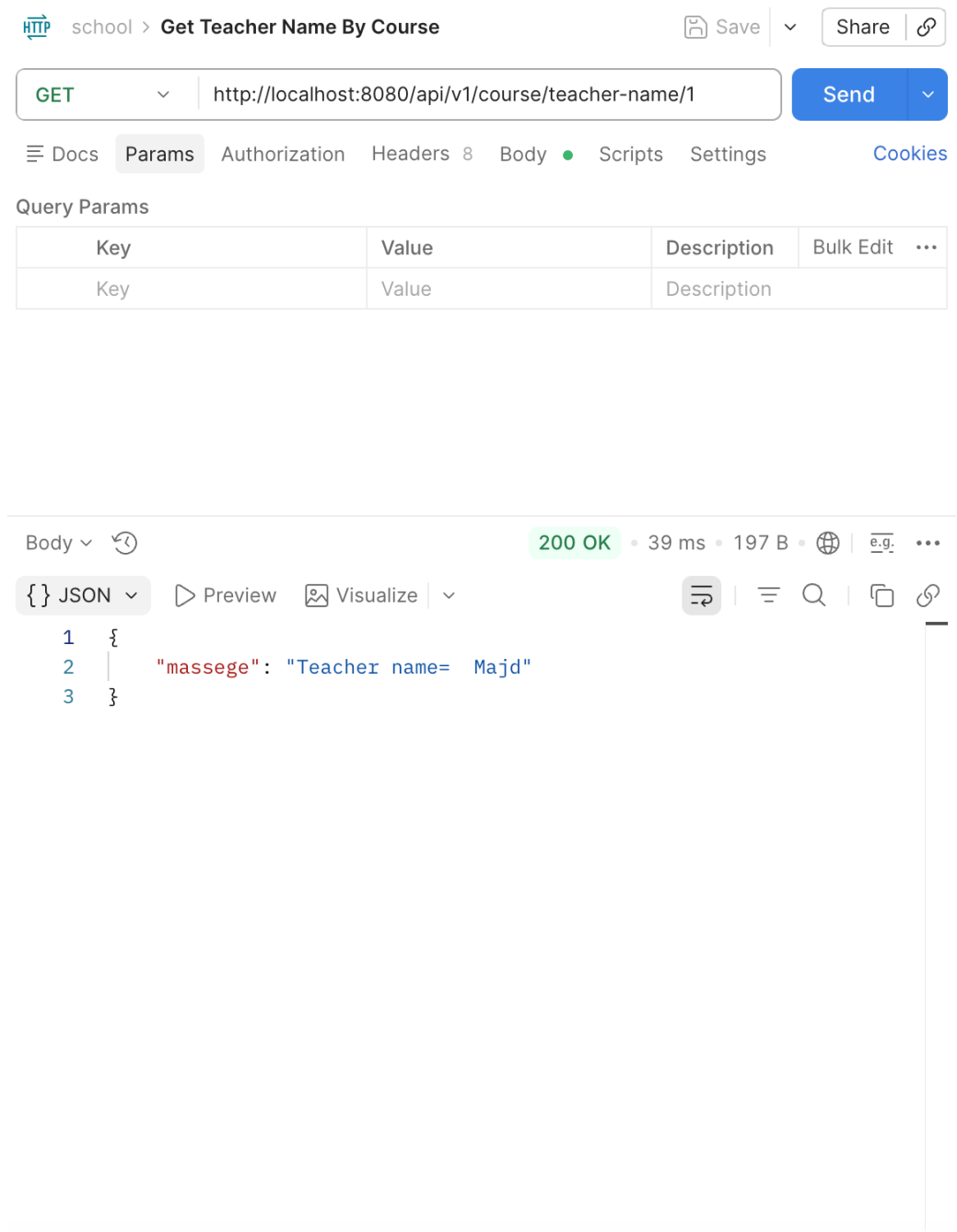


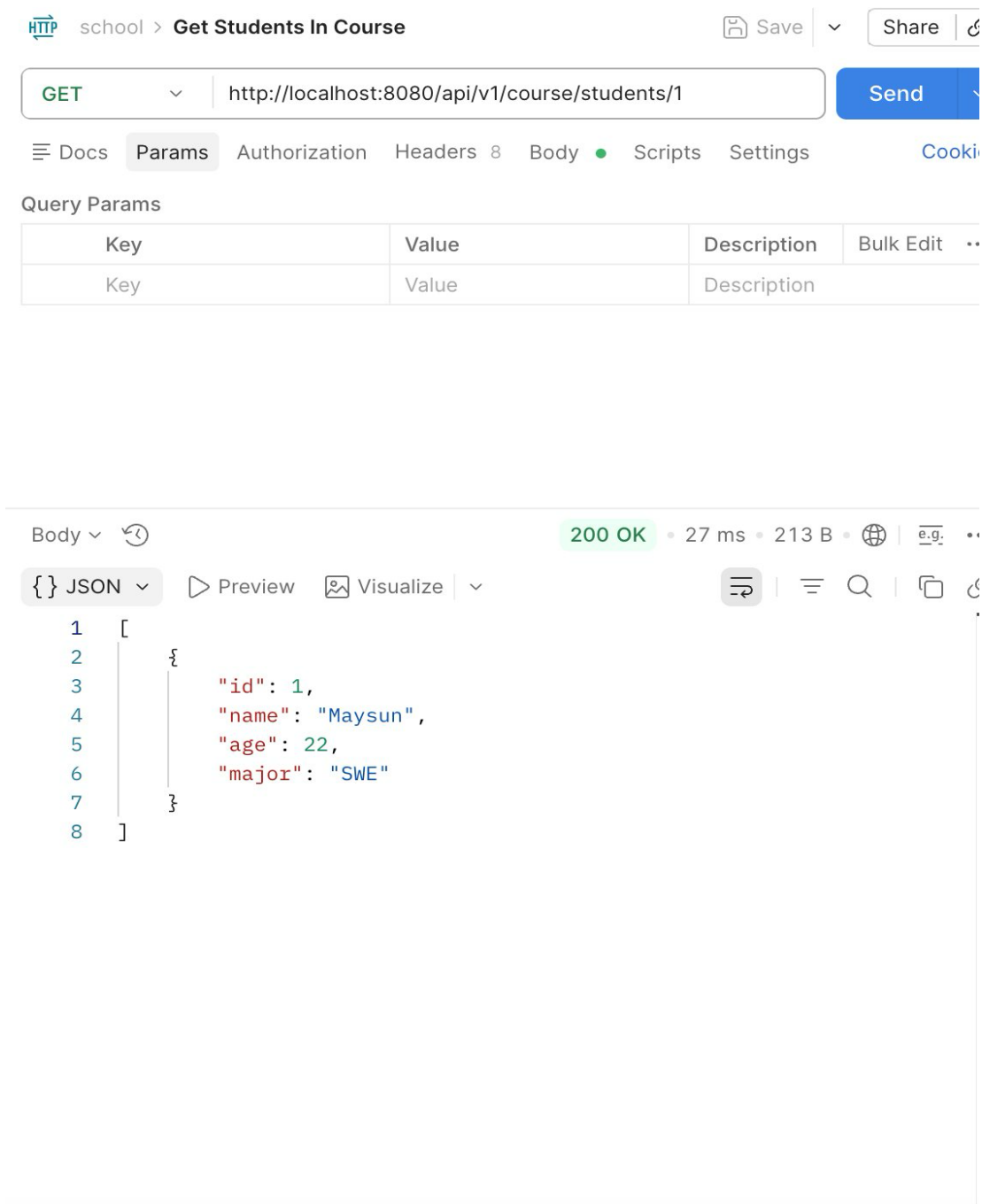
Figure 2 - Get Teacher Name By Course: returns teacher name

3.3 Get Students in Course

Endpoint:

GET /api/v1/course/students/1

Returns the list of students enrolled in a specific course



HTTP school > **Get Students In Course** Save Share

GET http://localhost:8080/api/v1/course/students/1 Send

Docs Params Authorization Headers 8 Body Scripts Settings Cookies

Query Params

Key	Value	Description	Bulk Edit
Key	Value	Description	

Body 200 OK • 27 ms • 213 B

{ } JSON Preview Visualize

```
1 [
2   {
3     "id": 1,
4     "name": "Maysun",
5     "age": 22,
6     "major": "SWE"
7   }
8 ]
```

Figure 3 - Get Students In Course: returns list

3.4 Change Student Major

Endpoint:

PUT /api/v1/student/change-major/1/Backend%20developer

Changes the student's major and automatically drops all courses the student was enrolled in, maintaining data consistency.

The screenshot displays a REST client interface for the endpoint `school > Change Student Major`. The request is a **PUT** to `http://localhost:8080/api/v1/student/change-major/1/Backend%20developer`. The **Params** tab is active, showing a table with one parameter:

Key	Value	Description	Bulk Edit	..
Key	Value	Description		

The response is a **200 OK** status with a response time of 201 ms and a body size of 211 B. The response body is shown in JSON format:

```
1 {
2   |   "massege": "major changed and courses dropped"
3   }
```

Figure 4 - Change Student Major

3.5 Verify Student Courses After Major Change

Endpoint:

GET /api/v1/student/get

After changing the major, the student list is retrieved to verify the new major is saved. The courses list is empty as expected after the major change.

The screenshot shows a REST client interface with the following components:

- URL Bar:** Method **GET**, URL `http://localhost:8080/api/v1/student/get`, and a **Send** button.
- Params Tab:** A table titled "Query Params" with columns "Key", "Value", "Description", and "Bulk Edit". It contains one row with "Key", "Value", "Description", and an empty "Bulk Edit" cell.
- Response Section:** Shows a status of **200 OK** with a response time of 43 ms and a size of 227 B. The response body is displayed in JSON format:

```
1  [
2    {
3      "id": 1,
4      "name": "Maysun",
5      "age": 22,
6      "major": "Backend developer"
7    }
8  ]
```

Figure 5 - new major shown, courses cleared